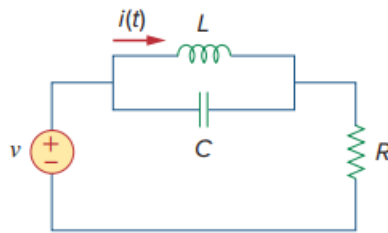


Problem

Using Fig. 8.93, design a problem to help other students better understand the step response of a parallel RLC circuit.

Figure 8.93:



Step-by-step solution

Step 1 of 9

Refer to Figure 8.93 in the textbook.

For $t < 0$, the circuit is in steady state, the capacitor is replaced with an open circuit and the inductor with a short circuit.

Re-draw the circuit diagram:

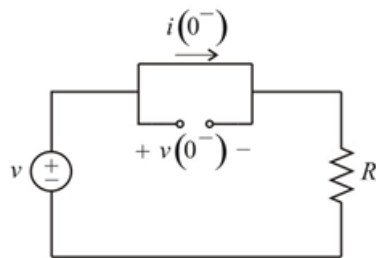


Figure 1

Step 2 of 9

From the above circuit, determine the initial conditions.

$$v(0^-) = 0 \text{ V}$$

$$i(0^-) = \frac{v}{R}$$

Step 3 of 9

Due to the continuity of the capacitor voltage and inductor current,

$$v(0^+) = v(0^-)$$

$$v(0^+) = 0 \text{ V}$$

$$i(0^+) = i(0^-)$$

$$i(0^+) = \frac{v}{R}$$

Step 4 of 9

For $t > 0$, the equivalent circuit is as shown below:

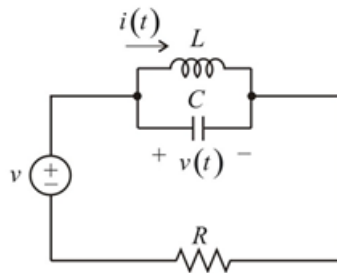


Figure 2

Step 5 of 9

Using source transformation technique, transform the voltage source v in series with the resistor R , into a current source $i = \frac{v}{R}$ in parallel with a resistor R .

Re-draw the circuit diagram:

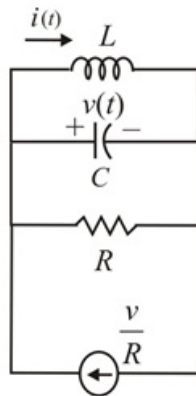


Figure 3

Step 6 of 9

It is a parallel RLC circuit with a current source,

$$\alpha = \frac{1}{2RC}$$

$$\omega_o = \frac{1}{\sqrt{LC}}$$

Hence, the roots are,

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_o^2}$$

Write the response of the circuit.

$$i(t) = i_t + i_{ss}$$

Where,

i_t is the transient response

i_{ss} is the steady state response

Step 7 of 9

There are three types of transient responses.

1. $\alpha > \omega_o$, overdamped response.

$$i_t = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

2. $\alpha = \omega_o$, critically damped response.

$$i_t = (A_1 + A_2 t) e^{s t}$$

3. $\alpha < \omega_o$, underdamped response.

$$i_t = (B_1 \cos \omega_d t + B_2 \sin \omega_d t) e^{-\alpha t}$$

Where,

$$\omega_d = \sqrt{\omega_o^2 - \alpha^2}$$

Step 8 of 9

Steady state response is the final value of the inductor current i.e., $i_{ss} = i(\infty)$.

Now as $t \rightarrow \infty$, the circuit reaches steady state and the inductor is replaced by a short circuit and the capacitor is replaced by an open circuit.

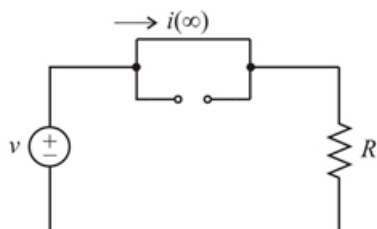


Figure 4

Step 9 of 9

From figure 4,

$$i(\infty) = \frac{v}{R}$$

$$i_{ss} = \frac{v}{R}$$

Hence, the final response is of the form,

$$i(t) = i_t + \frac{v}{R}$$

Where,

i_t is the transient response and the constants A_1, A_2, B_1 and B_2 are found by using the initial conditions.

Thus, step response of the system is explained.